ACMEGRADE

MINI PROJECT ON

IoT Based Smart Home Automation System Using Various Sensors

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MOOD INDIGO

IIT BOMBAY

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ABSTRACT

Modern homes require intelligent systems to improve safety, comfort, and efficient resource management. However, manual monitoring of appliances and environmental conditions can be inconvenient and inefficient. To address this problem, an IoT-based Smart Home Automation System is proposed. The system uses a microcontroller integrated with sensors such as soil moisture, PIR motion, ultrasonic distance, and gas sensors to monitor environmental conditions. Sensor data is processed and transmitted to the cloud using Wi-Fi connectivity for remote access. Users can monitor and control appliances through a mobile or web application. The system enhances home security, enables automatic control of devices, and supports efficient energy utilization. Overall, the project provides a cost-effective and scalable smart home solution.

1. INTRODUCTION

1.1. Motivation

The motivation of this project is to reduce human effort and improve safety through smart automation. Traditional home systems lack real-time monitoring and efficient resource management. By using IoT and sensors such as soil moisture, PIR, ultrasonic, and gas sensors, the system enables automatic control, security, and safety alerts. Remote monitoring through the internet increases convenience and energy efficiency in modern homes.

1.2. Proposed System

The proposed system is an **IoT-based Smart Home Automation System** designed to monitor environmental conditions and automatically control household appliances using multiple sensors and a microcontroller. An **Arduino Uno** acts as the central controller of the system.

The system uses a **soil moisture sensor** to monitor moisture levels and control water supply for household plants, preventing water wastage. A **PIR motion sensor** detects human movement and is used for security purposes and automatic lighting control. An **ultrasonic distance sensor** measures distance and enables applications such as automatic door operation or obstacle detection. A **gas sensor** continuously monitors the presence of harmful gases and activates a buzzer and alert system in case of gas leakage.

All sensor data is processed by the Arduino, which makes decisions based on predefined conditions. Appliances such as lights, bulbs, and alarms are controlled using **relay modules**. The system provides automatic operation, enhances safety, reduces manual effort, and improves energy efficiency. This proposed system is cost-effective, reliable, and suitable for modern smart home applications.

2. METHODOLOGY

2.1. Block Diagram

Figure 2.1. Block Diagram

The input section consists of various sensors that collect environmental data. The Arduino Uno processes the sensor data and makes decisions based on predefined conditions. The output section controls appliances and alerts through relays and buzzers, providing automation and safety.

2.2. Design Flow

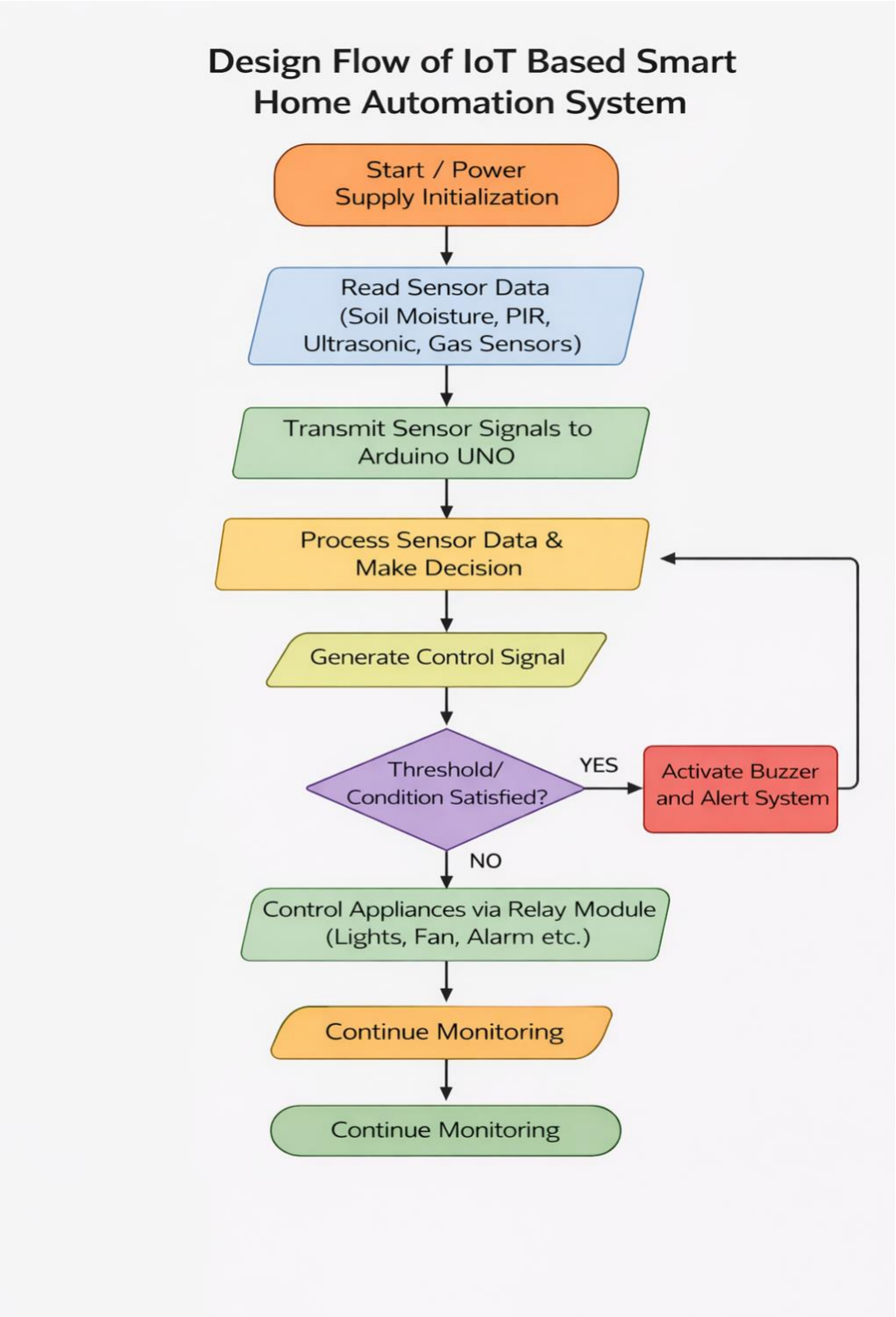
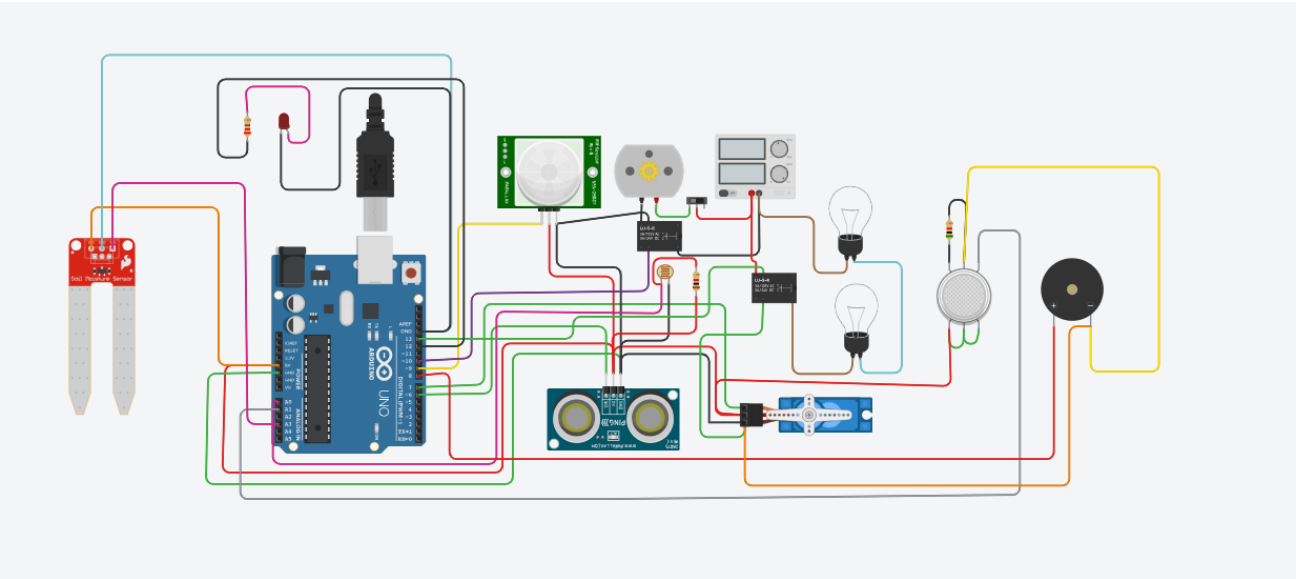


Figure 2.2. Design Flow

2.3. Circuit Diagram



Arduino Code

```
#include <Servo.h>

int output1Value = 0;
int sen1Value = 0;
int sen2Value = 0;

const int gas_sensor = A1;
const int LDR = A0;
int limit = 400;
int soilPin = A3;
int soilLed = 12;

long
readUltrasonicDistance(int
triggerPin, int echoPin)
{
  pinMode(triggerPin,
OUTPUT);
  digitalWrite(triggerPin,
LOW);
  delayMicroseconds(2);

  digitalWrite(triggerPin,
HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin,
LOW);

  pinMode(echoPin,
INPUT);
  return pulseIn(echoPin,
HIGH);
}

Servo servo_7;

void setup()
{
  Serial.begin(9600);

  pinMode(A0, INPUT);
  pinMode(A1, INPUT);
  pinMode(A3, INPUT);

  pinMode(13, OUTPUT);
  pinMode(12, OUTPUT);

  servo_7.attach(7, 500, 2500);

  pinMode(8, OUTPUT);
  pinMode(9, INPUT);
  pinMode(10, OUTPUT);

  pinMode(4, OUTPUT);
  pinMode(3, OUTPUT);
}

void loop()
{
  int val1 = analogRead(LDR);
  if (val1 > 500)
  {
    digitalWrite(13, LOW);
    Serial.print("Bulb ON = ");
```

```
    Serial.print(val1);
  }
  else
  {
    digitalWrite(13, HIGH);
    Serial.print("Bulb OFF =
");
    Serial.print(val1);
  }
  sen2Value =
digitalRead(9);
  if (sen2Value == 0)
  {
    digitalWrite(10, LOW);
    digitalWrite(4, HIGH);
    digitalWrite(3, LOW);
    Serial.print(" || NO
Motion Detected ");
  }

  if (sen2Value == 1)
  {
    digitalWrite(10, HIGH);
    delay(5000);
    digitalWrite(4, LOW);
    digitalWrite(3, HIGH);
    Serial.print(" || Motion
Detected! ");
  }
  int val =
analogRead(gas_sensor);
  Serial.print(" || Gas Sensor
Value = ");
  Serial.print(val);

  if (val > limit)
  {
    tone(8, 650);
  }
  delay(300);
  noTone(8);
  sen1Value = 0.01723 *
readUltrasonicDistance(6,
6);
  if (sen1Value < 100)
  {
    servo_7.write(90);
    Serial.print(" || Door
Open! Distance = ");
    Serial.print(sen1Value);
    Serial.print("\n");
  }
  Else
  {
    servo_7.write(0);
    Serial.print(" || Door Closed! Distance = ");
    Serial.print(sen1Value);
    Serial.print("\n");
  }
  int soilValue = analogRead(soilPin);
  Serial.print(" || Soil Moisture = ");
  Serial.print(soilValue);

  if (soilValue < 400)
  {
    digitalWrite(soilLed, HIGH);
  }
  else
  {
    digitalWrite(soilLed, LOW);
  }

  delay(10);
}
```

2.5 Limitations of Existing Systems

The existing home automation systems mainly rely on manual operation, which reduces convenience and efficiency. Most traditional systems do not provide remote access or internet-based control. They are often expensive and require complex wiring for installation. Scalability is limited, making it difficult to add new devices. These systems generally lack real-time alerts for emergencies such as gas leakage or intrusion. Energy consumption is higher due to the absence of automatic control based on environmental conditions. Additionally, security features in basic systems are limited and less reliable.

3. TOOLS AND LANGUAGES REQUIRED

3.1. Simulation Tool

TinkerCad

Tinkercad is a free online collection of software tools that help in the creation and designing of electronic circuits.

Figure 3.1.1. Tinkercad Software

3.2. Hardware Components

Arduino UNO

The Arduino Uno is a **microcontroller board** based on the **Microchip ATmega328P microcontroller** and developed by Arduino.cc. The board is equipped with digital and analog input/output (I/O) pins that may be interfaced to other circuits. The board has 14 digital I/O pins, 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable.

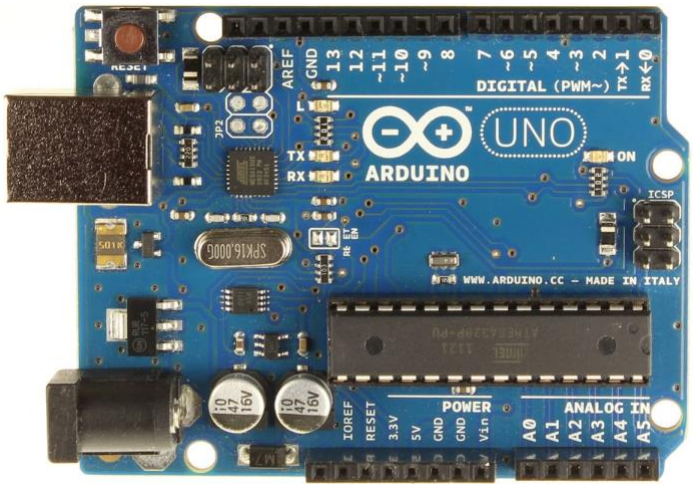


Figure 3.2.1. Arduino UNO Boar

Soil Moisture Sensor

Function:

Measures moisture level in soil. Used for automatic irrigation control.

Working Principle:

It measures resistance between probes.

Wet soil → low resistance → high moisture value.

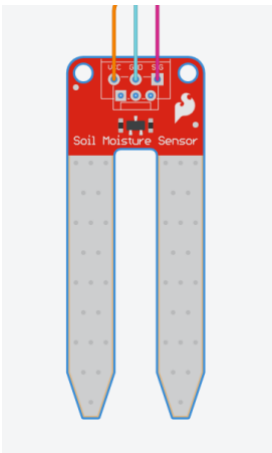


Figure 3.2.3. Soil Moisture Sensor

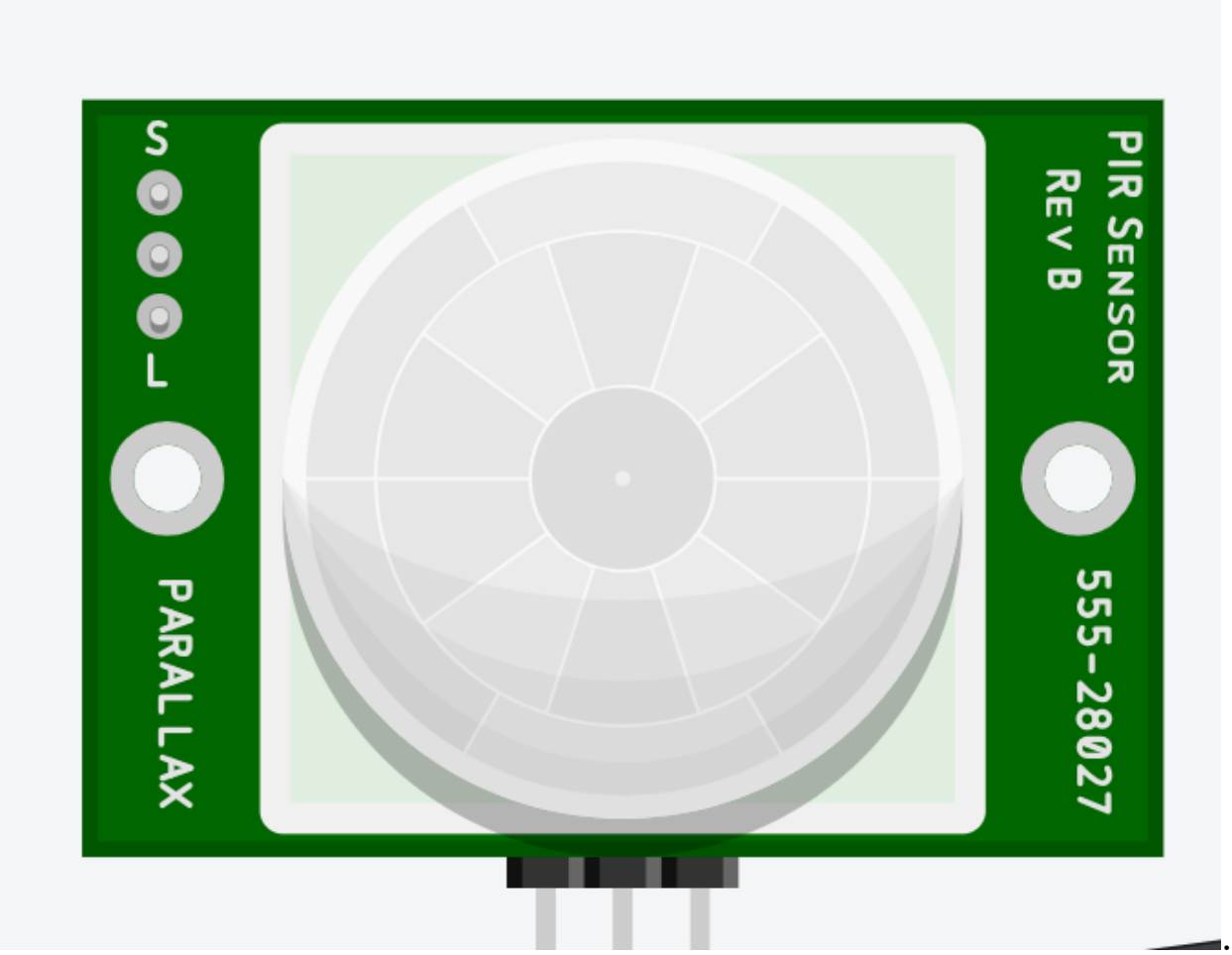
LED

A light-emitting diode (LED) is a semiconductor light source that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. The color of the light (corresponding to the energy of the photons) is determined by the energy required for electrons to cross the band gap of the semiconductor.

Figure 3.2.4. LEDs

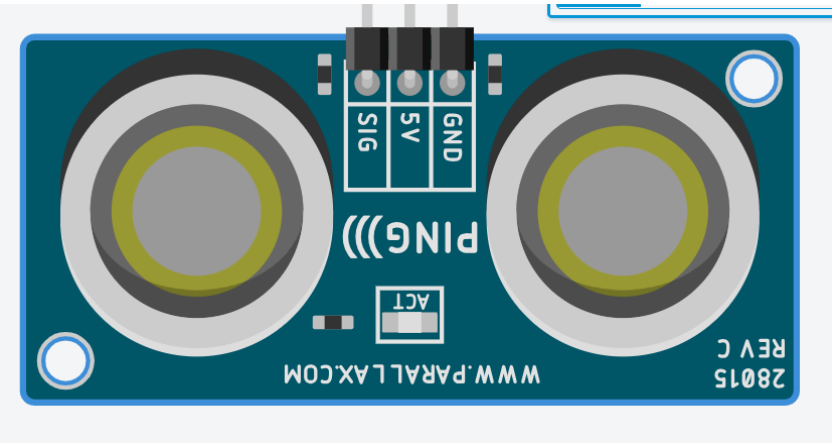
PIR Motion Sensor

HC-SR501
Function:
Detects human movement using infrared radiation.
Working:
When motion is detected, it sends HIGH signal to Arduino



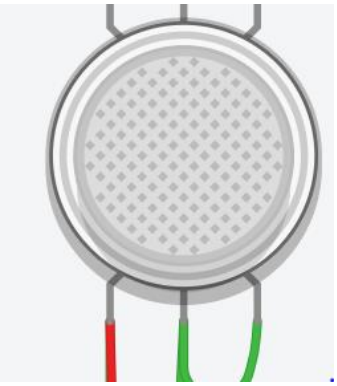
Ultrasonic Distance Sensor

HC-SR04
Function:
Measures distance using ultrasonic waves.
Working Formula:
Distance = (Time × Speed of Sound) / 2

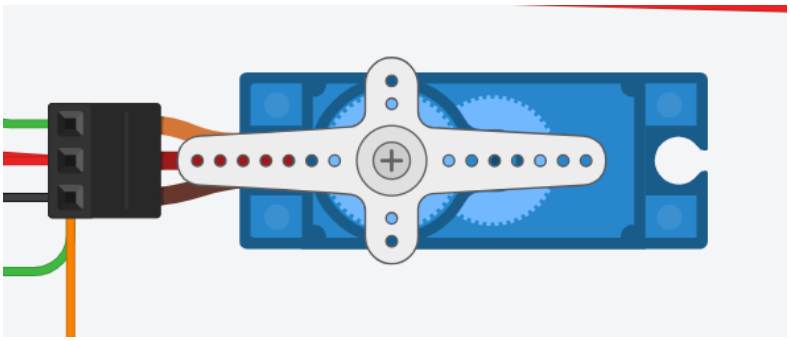


Gas Sensor

MQ-2
Function:
Detects LPG, methane, smoke, and flammable gases.
Output:
Analog voltage proportional to gas concentration.



Servo Motor



Servo Motor (SG90)

Working

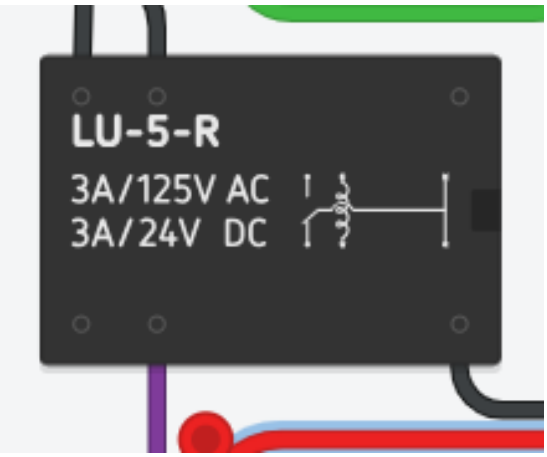
1. The servo motor operates using PWM (Pulse Width Modulation) signals from the microcontroller.
2. The width of the pulse determines the angle of rotation.
3. An internal DC motor rotates based on the received control signal.
4. A feedback mechanism ensures accurate positioning of the shaft.
5. The motor stops automatically when the desired angle is reached.

Output

1. The output of a servo motor is controlled angular movement.
2. It rotates within a fixed range of 0° to 180°.

3. The shaft position changes according to the PWM signal.
4. It provides precise and stable position control.
5. The output is mechanical rotational motion.

RELAY SPDT



Working: When the relay coil is energized, the common (COM) terminal switches from Normally Closed (NC) to Normally Open (NO)

Output: The device connected to NO turns ON when the relay is activated, and the device connected to NC turns ON when the relay is deactivated.

Input vs Output Table

Input Condition (Sensor)	Output Action (System Response)
Gas level > Threshold	Buzzer turns ON (Gas alert)
Soil moisture < 400 (Dry)	Soil LED turns ON
Object distance < 100 cm	Servo motor rotates (Door opens)
Darkness detected (Low light)	Bulb turns ON
Motion detected (PIR = 1)	Fan/Light turns ON + Green LED
No motion detected	Fan OFF + Red LED ON

4.ADVANTAGES

The proposed smart home automation system offers several important advantages compared to traditional systems.

1. **CompleteAutomation**
   The system automatically controls appliances based on sensor inputs such as motion detection, gas leakage, soil moisture level, and distance measurement. This reduces human effort and increases convenience.
2. **Energy Efficiency and Power Saving**
   Appliances operate only when required. For example, lights turn ON only when motion is detected, and irrigation works only when soil moisture is low. This reduces unnecessary power consumption and lowers electricity bills.
3. **Enhanced Safety and Security**
   The gas sensor detects harmful gases and activates a buzzer alert in case of leakage. The PIR sensor detects unauthorized movement. This improves home safety and prevents accidents.
4. **Cost-Effective Solution**
   The system uses affordable components such as the Arduino Uno and basic sensors, making it economical compared to commercial automation systems.
5. **Real-Time Monitoring**
   Sensors continuously monitor environmental conditions and send data to the controller instantly, ensuring immediate response to changes.
6. **Scalability and Flexibility**
   Additional sensors and modules can be easily added to expand the system. The design is flexible and can be modified according to user requirements.
7. **Low Power Consumption**
   The components used consume minimal power, making the system suitable for long-term operation.
8. **Easy Installation and Maintenance**
   The circuit design is simple, and components are easily available and replaceable. Maintenance is simple and does not require advanced technical skills.
9. **Improved Accuracy and Reliability**
   Sensors like ultrasonic and gas sensors provide accurate readings, ensuring

proper functioning of the system.

10. **User-Friendly Operation**
The system is easy to program and operate. Even beginners can understand and manage it.

5. FUTURE SCOPE

The proposed IoT Based Smart Home Automation System can be further enhanced with advanced technologies to improve performance, intelligence, and user convenience.

1. **Integration with IoT and Cloud Platforms**
   The system can be connected to cloud servers for remote monitoring and control through mobile applications from anywhere in the world.
2. **Artificial Intelligence Integration**
   AI algorithms can be added to analyze user behavior and automatically adjust appliances based on usage patterns.
3. **Voice Control System**
   The project can be integrated with voice assistants like Amazon Alexa or Google Assistant for hands-free control.
4. **Mobile Application Development**
   A dedicated Android or iOS app can be developed to monitor sensor values and control appliances easily.
5. **Advanced Security Features**
   Camera modules, fingerprint sensors, or face recognition systems can be added for better home security.
6. **Energy Monitoring System**
   Smart energy meters can be integrated to monitor power consumption and optimize energy usage.
7. **Wireless Communication Upgrade**
   The system can be upgraded using Wi-Fi, Bluetooth, or Zigbee modules for better connectivity and flexibility.
8. **Smart City Integration**
   The system can be expanded and connected with smart grid and smart city infrastructure in the future.

CONCLUSION

The **IoT Based Smart Home Automation System Using Various Sensors** successfully demonstrates an efficient and intelligent method for automating household appliances. The system integrates multiple sensors such as soil moisture sensor, PIR motion sensor, ultrasonic sensor, and gas sensor with a microcontroller to monitor environmental conditions and control devices automatically.

The project improves convenience by reducing manual operation and enhances safety through real-time detection of gas leakage and unauthorized movement. It also promotes energy efficiency by operating appliances only when required. The use of affordable components makes the system cost-effective and suitable for residential applications.

Overall, the proposed system provides a reliable, scalable, and practical solution for modern smart homes and lays the foundation for future advancements in IoT-based automation technologi